Women Safety Companion

A Smart Web Application for Women's Personal Safety

Live Demo: https://safe-guard-companion.base44.app

Project Documentation

Project Type: B.Tech CSE Mini Project

Development Tool: Base44 AI

Application Category: Web Application / Women Safety / Social Impact

1. Project Overview

Women Safety Companion is a web-based safety application designed to provide women with quick access to essential safety tools during normal travel and emergency situations. The application provides emergency contacts, an SOS workflow, safe-place information, safety tips, incident reporting, and safety checklists through a simple and responsive interface.

This project was developed as a B.Tech CSE Mini Project using Base44 AI, which helped in designing and developing the web application through AI-assisted application development.

2. Problem Statement

Women may face unsafe situations while travelling, especially in unfamiliar places or during emergencies. In such situations, finding emergency contacts, identifying nearby safe places, and accessing safety information quickly can be difficult. Therefore, Women Safety Companion aims to provide these essential safety features through a single web application.

3. Proposed Solution

- Quickly access emergency contacts.
- Trigger an SOS workflow.
- Find nearby safe places.
- View emergency and safety information.
- Report safety incidents.
- Follow personal safety checklists.
- Access practical safety tips.

4. Key Features

4.1 SOS Emergency Workflow

The application provides a prominent SOS button. When activated, a short cancellation countdown is displayed. The user can cancel an accidental alert. After confirmation, an emergency screen displays emergency contacts and available emergency-service options, and an incident record can be created.

Prototype note: Emergency calls/messages are simulated unless a real emergency API or service is connected.

4.2 Emergency Contacts

- Add emergency contacts
- Edit and delete contacts
- Mark primary contacts
- View relationship and phone number
- Quick call option

4.3 Safe Places

The application provides information about potentially useful nearby locations such as police stations, hospitals, pharmacies, public help centers, and other predefined safe locations. Demo/sample locations can be used when live map services are not connected.

4.4 Incident Reporting

Users can report safety-related incidents using incident type, location, date/time, description, and optional image information.

4.5 Safety Status

The application provides an informational safety-status concept using Safe, Moderate Risk, and High Risk categories. This status does not guarantee actual safety.

4.6 Safety Checklist

- Emergency contacts added
- Phone charged
- Location services enabled
- Emergency numbers saved
- Safe route checked

4.7 Safety Tips

The application provides practical guidance covering travel safety, night-time safety, public transportation, online safety, and emergency preparation.

5. System Architecture

User → Women Safety Companion Web App → Emergency Contacts / Safe Places / Incident Reporting → Base44 Application & Data Management

6. Technologies Used

Technology	Purpose
Base44 AI	AI-assisted application development
React	Frontend application
HTML	Web page structure
CSS / Tailwind-style UI	Styling and responsive design
JavaScript	Application functionality
Base44 Backend	Data and application functionality
Web Browser	Application access

7. Role of Base44 AI

Base44 AI was used as the primary AI-assisted development platform for creating the application. It helped with application structure, UI design, page creation, component generation, data models, user-interface improvements, application workflows, responsive design, and feature implementation.

8. Main Application Modules

User Authentication → Dashboard → Emergency Contacts → Safe Places → Incident Reporting → Safety Tips → User Profile

9. User Workflow

Start → Register/Login → Safety Dashboard → Select Required Feature → SOS / Emergency Contacts / Safe Places / Incident Reporting → Emergency Workflow or Information Display

10. User Interface

The application follows a clean and responsive design suitable for desktop and mobile devices. The interface focuses on simple navigation, a large SOS button, clear emergency actions, high visibility, responsive layouts, easy-to-understand cards, and accessible controls.

11. Privacy & Safety

- User information should not be publicly exposed.
- Emergency contacts are associated with the respective user.
- Sensitive information should be handled securely.
- The application does not guarantee physical safety.
- Emergency features should be connected to verified emergency services before real-world deployment.

12. Limitations

- Real-time GPS functionality may require additional location services.

- Real SMS/call automation requires appropriate APIs and permissions.
- Safe-place information may use demo data.
- The safety score is informational and cannot accurately predict real-world danger.
- Integration with police/emergency services is not included in the prototype.
- Internet connectivity may be required for some features.

13. Future Enhancements

- Real-time GPS tracking
- Google Maps/OpenStreetMap integration
- Real-time SOS alerts
- SMS alerts to emergency contacts
- One-tap emergency calling
- AI-based safety assistant
- Voice-activated SOS
- AI-based risk analysis
- Real-time route safety analysis
- Safety incident analytics
- Integration with nearby hospitals and police stations
- Progressive Web App / mobile application
- Advanced authentication and encryption

14. Expected Outcomes

- Provide quick access to emergency resources.
- Improve emergency preparedness.
- Help users manage trusted emergency contacts.
- Provide information about potentially useful safe places.
- Encourage users to follow safety practices.
- Demonstrate the use of AI-assisted development in solving a real-world problem.

15. Academic Information

Project Title: Women Safety Companion

Student: Chandrika Rachamreddy

Program: B.Tech – Computer Science and Engineering

Project Type: Mini Project

Development Tool: Base44 AI

16. Live Application

https://safe-guard-companion.base44.app

17. Disclaimer

This project is developed for educational and demonstration purposes as a B.Tech CSE mini project. The application should not be considered a replacement for official emergency services. In a real emergency, users should contact the appropriate local emergency services immediately.

18. Conclusion

Women Safety Companion demonstrates how AI-assisted development tools such as Base44 AI can be used to build a practical web application addressing a real-world social problem. The project combines web development, data management, user workflows, emergency-support features, and responsive UI design into a single platform.